JSS 3 Web Development – Full Notes + HTML & CSS Projects

Note: Please, you need to use External CSS Styling

Week 1: Forms & Inputs

Notes

- **Forms** are used to collect user input (like login, registration, surveys).
- The <form> tag wraps input fields.
- Common <input> types:
    - text → plain text
    - password → hidden text
    - email → email input
    - number → numbers only
    - radio → choose one option
    - checkbox → choose multiple options
    - submit → button to send data

Real-Time Project: Student Login Page

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <title>Student Login</title>
```

```
<style>
  body {
    font-family: Arial, sans-serif;
    background: #eef2f7;
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
  }
  .login-box {
    background: white;
    padding: 30px;
    border-radius: 10px;
    box-shadow: 0 4px 10px rgba(0,0,0,0.2);
    width: 320px;
  }
  .login-box h2 {
    text-align: center;
    margin-bottom: 20px;
    color: #003366;
  }
  label {
    font-weight: bold;
    display: block;
    margin: 8px 0 4px;
  }
```

```
    input[type="text"], input[type="password"] {
      width: 100%;
      padding: 10px;
      border: 1px solid #ccc;
      border-radius: 6px;
      margin-bottom: 15px;
    }
    input[type="submit"] {
      width: 100%;
      padding: 10px;
      background: #003366;
      color: white;
      border: none;
      border-radius: 6px;
      cursor: pointer;
    }
    input[type="submit"]:hover {
      background: #00509e;
    }
  </style>
</head>
<body>
  <div class="login-box">
    <h2>Student Login</h2>
    <form>
      <label>Username:</label>
```

```
    <input type="text" name="username" placeholder="Enter username" required>

    <label>Password:</label>
    <input type="password" name="password" placeholder="Enter password" required>

    <input type="submit" value="Login">
  </form>
 </div>
</body>
</html>
```

Explanation:

- We styled the login box with box-shadow for a card effect.
- flexbox centers the form on the page.
- Username & password fields are styled with rounded borders.

Week 2: Registration Form

Notes

- Registration forms usually collect more data.
- Additional tags:
    - <label> → description for inputs
    - <textarea> → multi-line text
    - <select> + <option> → dropdown menus
    - <button> → clickable button

Real-Time Project: Student Registration Form

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <title>Student Registration</title>
 <style>
  body {
   font-family: Verdana, sans-serif;
   background: #f4f6f9;
   padding: 40px;
  }
  .form-container {
   background: #fff;
   padding: 25px;
   border-radius: 12px;
```

```
  max-width: 500px;
  margin: auto;
  box-shadow: 0 4px 12px rgba(0,0,0,0.2);
}
h2 {
  text-align: center;
  color: #003366;
  margin-bottom: 20px;
}
label {
  display: block;
  margin: 10px 0 5px;
  font-weight: bold;
}
input, select, textarea, button {
  width: 100%;
  padding: 10px;
  margin-bottom: 15px;
  border: 1px solid #bbb;
  border-radius: 6px;
}
button {
  background: #28a745;
  color: white;
  font-size: 16px;
  border: none;
```

```
    cursor: pointer;
   }
   button:hover {
    background: #218838;
   }
  </style>
</head>
<body>
  <div class="form-container">
   <h2>Student Registration</h2>
   <form>
    <label>Full Name:</label>
    <input type="text" name="fullname" placeholder="Enter full name" required>

    <label>Email:</label>
    <input type="email" name="email" placeholder="Enter email" required>

    <label>Age:</label>
    <input type="number" name="age" min="10" max="20" required>

    <label>Gender:</label>
    <input type="radio" name="gender" value="Male"> Male
    <input type="radio" name="gender" value="Female"> Female

    <label>Class:</label>
    <select name="class">
```

```
      <option>JSS 1</option>
      <option>JSS 2</option>
      <option>JSS 3</option>
    </select>

    <label>Short Bio:</label>
    <textarea name="bio" rows="4" placeholder="Tell us about yourself"></textarea>

    <button type="submit">Register</button>
  </form>
 </div>
</body>
</html>
```

Explanation:

- We used max-width: 500px; margin: auto; to center the form.
- Dropdown menu allows class selection.
- Radio buttons let users choose one gender.

Week 3: Semantic HTML Tags (Homepage)

Notes

- **Semantic tags** give meaning to a webpage structure.
- <header> → top section with logo & navigation
- <nav> → links for moving around site
- <main> → main content area
- <section> → groups related content
- <article> → independent content (like a news post)
- <aside> → side info (like events or ads)
- <footer> → bottom info (contacts, copyright)

Real-Time Project: School Homepage

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <title>My School Website</title>
 <style>
  body {
   font-family: Arial, sans-serif;
   margin: 0;
   background: #f9f9f9;
  }
  header {
   background: #003366;
```

```
  color: white;
  padding: 20px;
  text-align: center;
}
nav a {
  color: white;
  margin: 0 15px;
  text-decoration: none;
  font-weight: bold;
}
nav a:hover {
  text-decoration: underline;
}
main {
  display: flex;
  padding: 20px;
}
section {
  flex: 2;
  margin-right: 20px;
}
article {
  background: #eaf2f7;
  padding: 15px;
  border-radius: 10px;
  margin-top: 15px;
```

```
    }
    aside {
      flex: 1;
      background: #fff3cd;
      padding: 15px;
      border-radius: 10px;
    }
    footer {
      text-align: center;
      background: #003366;
      color: white;
      padding: 15px;
      margin-top: 20px;
    }
  </style>
</head>
<body>
  <header>
    <h1>My School</h1>
    <nav>
      <a href="#">Home</a>
      <a href="#">About</a>
      <a href="#">News</a>
      <a href="#">Contact</a>
    </nav>
  </header>
```

```
 <main>
  <section>
   <h2>Welcome to Our School</h2>
   <p>We provide excellent education and extracurricular activities.</p>

   <article>
    <h3>Latest News</h3>
    <p>Our JSS 3 students won the inter-school science competition!</p>
   </article>
  </section>

  <aside>
   <h3>Upcoming Events</h3>
   <ul>
    <li>Sports Day – Sept 20</li>
    <li>Cultural Day – Oct 1</li>
    <li>Graduation – Nov 25</li>
   </ul>
  </aside>
 </main>

 <footer>
  <p>&copy; 2025 My School. All Rights Reserved.</p>
 </footer>
</body>
```

</html>

Explanation:

- Used **Flexbox** (display: flex;) to arrange section and aside side by side.
- article highlights news with background color.
- header and footer styled with a blue theme.

Week 4: Practice – School Information Website

Note

In this week, students will bring together what they have learned in the last 3 weeks:

1. **Forms and Inputs** → Used to register students.
2. **Semantic Tags** → Used to structure the website into meaningful sections (header, nav, main, section, footer).
3. **Multimedia** → Add school anthem (audio), school tour video, and Google Map (iframe).

The goal is to design a **School Information Website** that contains:

- A **Home Page** with school info.
- A **Registration Form** for new students.
- A **Media Section** with video, audio, and map.
- A **Footer** with copyright.

Full HTML + CSS Code

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <title>School Information Website</title>
 <style>
  body {
   font-family: Arial, sans-serif;
   margin: 0;
```

```
  padding: 0;
  background: #f9f9f9;
}

header {
  background: navy;
  color: white;
  padding: 20px;
  text-align: center;
}

nav {
  background: #444;
  padding: 10px;
  text-align: center;
}

nav a {
  color: white;
  text-decoration: none;
  margin: 0 15px;
}

nav a:hover {
  text-decoration: underline;
}
```

```
main {
  padding: 20px;
}

section {
  margin-bottom: 30px;
  padding: 20px;
  background: white;
  border-radius: 8px;
  box-shadow: 2px 2px 6px rgba(0,0,0,0.1);
}

form label {
  display: block;
  margin-top: 10px;
  font-weight: bold;
}

form input, form button {
  width: 100%;
  padding: 8px;
  margin-top: 5px;
}

form button {
```

```
    background: navy;
    color: white;
    border: none;
    margin-top: 15px;
    cursor: pointer;
  }

  form button:hover {
    background: darkblue;
  }

  footer {
    background: #222;
    color: white;
    text-align: center;
    padding: 15px;
  }

  iframe, video, audio {
    display: block;
    margin: 15px auto;
    max-width: 100%;
  }
 </style>
</head>
<body>
```

```
<!-- HEADER -->
<header>
  <h1>Welcome to Bright Future Secondary School</h1>
  <p>Raising Future Leaders Through Education</p>
</header>

<!-- NAVIGATION -->
<nav>
  <a href="#about">About</a>
  <a href="#registration">Registration</a>
  <a href="#media">Media</a>
  <a href="#contact">Contact</a>
</nav>

<!-- MAIN CONTENT -->
<main>
  <!-- About Section -->
  <section id="about">
    <h2>About Our School</h2>
    <p>Bright Future Secondary School is committed to excellence in academics, morals, and skills development.
      We provide a conducive environment for students to thrive and achieve their dreams.</p>
  </section>

  <!-- Registration Form -->
  <section id="registration">
```

```
 <h2>Student Registration</h2>
 <form>
  <label for="name">Full Name:</label>
  <input type="text" id="name" placeholder="Enter your full name" required>

  <label for="email">Email Address:</label>
  <input type="email" id="email" placeholder="Enter your email" required>

  <label for="dob">Date of Birth:</label>
  <input type="date" id="dob">

  <label>Gender:</label>
  <input type="radio" name="gender" value="Male"> Male
  <input type="radio" name="gender" value="Female"> Female

  <label for="passport">Upload Passport:</label>
  <input type="file" id="passport">

  <button type="submit">Register</button>
 </form>
</section>

<!-- Media Section -->
<section id="media">
 <h2>School Media</h2>
```

```
    <h3>School Anthem</h3>
    <audio controls>
      <source src="anthem.mp3" type="audio/mpeg">
      Your browser does not support audio.
    </audio>

    <h3> School Tour Video</h3>
    <video controls width="400">
      <source src="schooltour.mp4" type="video/mp4">
      Your browser does not support video.
    </video>

    <h3>Our Location</h3>
    <iframe src="https://maps.google.com" width="400" height="300"></iframe>
  </section>
 </main>

 <!-- FOOTER -->
 <footer>
  <p>&copy; 2025 Bright Future Secondary School. All Rights Reserved.</p>
 </footer>
</body>
</html>
```

Real-Time Project

School Information Website features:

- **About Section** → Brief info about school.
- **Student Registration Form** → Collects student details.
- **Media Section** → Includes school anthem (audio), school tour (video), and school location (map).
- **Navigation** → Easy movement between sections.
- **Footer** → Copyright.

Week 5: CSS Syntax & Selectors

Note:

- Syntax: selector { property: value; }
- Selectors:
    - Tag selector: p { }
    - Class selector: .class { }
    - ID selector: #id { }
    - Group selector: h1, h2 { }

Code Example:

```
<!DOCTYPE html>
<html>
<head>
<style>
 body { background-color: lightblue; }
 h1 { color: navy; }
 .highlight { color: red; font-weight: bold; }
 #unique { font-size: 20px; }
</style>
</head>
<body>
 <h1 id="unique">Welcome to CSS</h1>
 <p class="highlight">This is important text.</p>
</body>
</html>
```

Real-Time Project: Personal Bio Page styled with selectors.

Week 6: Box Model, Colors, Units

Note:

- Box Model: Content → Padding → Border → Margin.
- Units: px, em, %, rem, vh/vw.
- Colors: names, HEX (#ff0000), RGB, HSL.

Code Example:

```
<!DOCTYPE html>
<html>
<head>
<style>
 .card {
   width: 300px;
   padding: 20px;
   border: 2px solid black;
   margin: 20px auto;
   background-color: #f9f9f9;
 }
</style>
</head>
<body>
 <div class="card">
   <h2>Student Profile</h2>
   <p>Name: John Doe</p>
   <p>Class: JSS 3</p>
 </div>
</body>
```

</html>

Real-Time Project:

Student Profile Card using box model**.**

Week 7: Fonts, Borders, Shadows

Note:

- Fonts: font-family, font-size, font-style, font-weight.
- Borders: style, color, radius.
- Shadows: box-shadow, text-shadow.

Code Example:

```
<!DOCTYPE html>
<html>
<head>
<style>
 .card {
   font-family: Arial;
   border: 2px solid blue;
   border-radius: 10px;
   box-shadow: 2px 2px 8px gray;
   padding: 20px;
 }
 h2 {
   text-shadow: 1px 1px 3px gray;
 }
</style>
</head>
<body>
 <div class="card">
   <h2>School Library</h2>
   <p>Welcome to our modern school library.</p>
```

```
  </div>
</body>
</html>
```

Real-Time Project:

School Library Info Card with shadows.

Week 8: Practice – Student Dashboard Page:

NOTE (External CSS)

Note

A **dashboard** is a user interface that shows key information in one place.
In this project, students will create a **Student Dashboard Page** containing:

1. **Student Bio Card** – Displays student profile info.
2. **Courses Card** – Shows enrolled subjects/courses.
3. **Notices Card** – Displays latest school notices.

We will style the cards with:

- **Box Model** (padding, margin, border).
- **Colors** (backgrounds and text).
- **Fonts & Shadows** for beauty.

Full HTML + CSS Code

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <title>Student Dashboard</title>
 <style>
  body {
   font-family: Arial, sans-serif;
   margin: 0;
   background: #f4f4f9;
```

```
}

header {
  background: navy;
  color: white;
  text-align: center;
  padding: 20px;
}

.dashboard {
  display: flex;
  justify-content: space-around;
  margin: 30px;
  flex-wrap: wrap;
}

.card {
  background: white;
  border-radius: 12px;
  box-shadow: 2px 2px 8px rgba(0,0,0,0.2);
  width: 300px;
  padding: 20px;
  margin: 15px;
}

.card h2 {
```

```
    color: navy;
    text-shadow: 1px 1px 3px lightgray;
   }

   .profile-pic {
    width: 100px;
    height: 100px;
    border-radius: 50%;
    border: 3px solid navy;
    display: block;
    margin: 0 auto 15px auto;
   }

   ul {
    list-style-type: square;
    padding-left: 20px;
   }

   .notice {
    background: #ffefc2;
    border-left: 5px solid orange;
    padding: 10px;
    margin-bottom: 10px;
   }
  </style>
</head>
```

```
<body>
 <!-- HEADER -->
 <header>
  <h1>Student Dashboard</h1>
  <p>Welcome back, John Doe</p>
 </header>

 <!-- DASHBOARD CARDS -->
 <div class="dashboard">
  <!-- Student Bio -->
  <div class="card">
   <h2> 👤 Student Bio</h2>
   <img src="student.jpg" alt="Student Picture" class="profile-pic">
   <p><b>Name:</b> John Doe</p>
   <p><b>Class:</b> JSS 3</p>
   <p><b>Email:</b> johndoe@email.com</p>
  </div>

  <!-- Courses -->
  <div class="card">
   <h2> 📚 Courses</h2>
   <ul>
    <li>Mathematics</li>
    <li>English Language</li>
    <li>Basic Science</li>
    <li>Computer Studies</li>
```

```
    </ul>
  </div>

  <!-- Notices -->
  <div class="card">
    <h2>📌 Notices</h2>
    <div class="notice">
      <p><b>📌</b> Mid-term exams start next week.</p>
    </div>
    <div class="notice">
      <p><b>📌</b> School sports day is on Friday.</p>
    </div>
    <div class="notice">
      <p><b>📌</b> Library books must be returned by Monday.</p>
    </div>
  </div>
 </div>
</body>
</html>
```

Real-Time Project

The **Student Dashboard Page** includes:

- **Bio Card** → Student name, class, email, and profile picture.
- **Courses Card** → List of enrolled subjects.
- **Notices Card** → School announcements.
- **Styling** → Cards with shadows, rounded borders, and color highlights.

Week 9: Display & Positioning

📖 **Note:**

- Display: block, inline, inline-block, none.
- Position: static, relative, absolute, fixed, sticky.

💻 **Code Example:**

```
<!DOCTYPE html>
<html>
<head>
<style>
 .nav {
   background: navy;
   color: white;
   padding: 10px;
   position: fixed;
   top: 0;
   width: 100%;
 }
</style>
</head>
<body>
 <div class="nav">School Navigation Bar</div>
 <p style="margin-top:60px">Page Content Starts Here...</p>
</body>
</html>
```

🎯 **Real-Time Project:**

Fixed School Navigation Bar.

Week 10: Flexbox

Note:

- Flexbox arranges elements in rows or columns.
- Properties: display:flex, justify-content, align-items.

Code Example:

```
<!DOCTYPE html>
<html>
<head>
<style>
 .container {
   display: flex;
   justify-content: space-around;
   margin-top: 30px;
 }
 .box {
   background: lightgreen;
   padding: 20px;
   width: 100px;
 }
</style>
</head>
<body>
 <h2>Student Subjects</h2>
 <div class="container">
   <div class="box">Math</div>
   <div class="box">Science</div>
```

```
  <div class="box">English</div>
 </div>
</body>
</html>
```

Real-Time Project:

Subjects Layout Page with Flexbox.

Week 11: CSS Grid

Note:

- Grid creates rows and columns.
- display:grid, grid-template-columns, gap.

Code Example:

```
<!DOCTYPE html>
<html>
<head>
<style>
 .grid {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: 10px;
 }
 .box {
  background: lightcoral;
  padding: 20px;
  text-align: center;
 }
```

```
</style>
</head>
<body>
  <h2>School Timetable</h2>
  <div class="grid">
    <div class="box">Math</div>
    <div class="box">English</div>
    <div class="box">Science</div>
    <div class="box">History</div>
    <div class="box">Art</div>
    <div class="box">PE</div>
  </div>
</body>
</html>
```

Real-Time Project:

School Timetable Layout with CSS Grid.

Week 12: Practice

Project: Full School Website Layout using Flexbox & Grid.

Note: External CSS

- **Flexbox → Best for row/column alignment and distributing space.**
- **Grid → Best for page layout with rows & columns.**

In this practice, we'll design a School Website Layout with:

1. **Header → School name + navigation (Flexbox).**
2. **Hero Section → Welcome message (Flexbox).**
3. **Main Content → Grid with 3 sections: About, Courses, and Notices.**
4. **Footer → Contact info.**

This project combines semantic structure + flexbox + grid.

```
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <title>School Website Layout</title>
 <style>
  body {
   margin: 0;
   font-family: Arial, sans-serif;
   background: #f4f4f9;
  }

  /* HEADER */
```

```
header {
  background: navy;
  color: white;
  padding: 20px;
}

nav {
  display: flex;
  justify-content: space-between;
  align-items: center;
}

nav .logo {
  font-size: 22px;
  font-weight: bold;
}

nav ul {
  list-style: none;
  display: flex;
  margin: 0;
  padding: 0;
}

nav ul li {
  margin-left: 20px;
```

```
}

nav ul li a {
  color: white;
  text-decoration: none;
}

nav ul li a:hover {
  text-decoration: underline;
}

/* HERO SECTION */
.hero {
  display: flex;
  justify-content: center;
  align-items: center;
  background: url("school.jpg") no-repeat center/cover;
  color: white;
  text-align: center;
  height: 250px;
}

.hero h1 {
  background: rgba(0,0,0,0.5);
  padding: 20px;
  border-radius: 10px;
```

```
}

/* MAIN GRID */
main {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: 20px;
  padding: 20px;
}

.card {
  background: white;
  border-radius: 8px;
  box-shadow: 2px 2px 8px rgba(0,0,0,0.2);
  padding: 20px;
}

.card h2 {
  color: navy;
}

/* FOOTER */
footer {
  background: #222;
  color: white;
  text-align: center;
```

```
    padding: 15px;
    margin-top: 20px;
  }

  /* RESPONSIVE DESIGN */
  @media (max-width: 768px) {
    main {
      grid-template-columns: 1fr;
    }

    .hero {
      height: 180px;
    }
  }
 </style>
</head>
<body>
 <!-- HEADER -->
 <header>
  <nav>
   <div class="logo">Bright Future School</div>
   <ul>
    <li><a href="#">Home</a></li>
    <li><a href="#">About</a></li>
    <li><a href="#">Courses</a></li>
    <li><a href="#">Notices</a></li>
```

```
      <li><a href="#">Contact</a></li>
    </ul>
  </nav>
 </header>

 <!-- HERO SECTION -->
 <section class="hero">
  <h1>Welcome to Bright Future School</h1>
 </section>

 <!-- MAIN GRID -->
 <main>
  <div class="card">
    <h2>🏫 About Us</h2>
    <p>We are committed to academic excellence and holistic student development. Our teachers are dedicated to nurturing future leaders.</p>
  </div>
  <div class="card">
    <h2>📚 Courses</h2>
    <ul>
      <li>Mathematics</li>
      <li>English Language</li>
      <li>Basic Science</li>
      <li>Computer Studies</li>
    </ul>
  </div>
```

```
  <div class="card">
    <h2>📌 Notices</h2>
    <p>📌 Mid-term exams start next week.</p>
    <p>📌 Sports Day is on Friday.</p>
    <p>📌 Library books must be returned by Monday.</p>
  </div>
 </main>

 <!-- FOOTER -->
 <footer>
  <p>&copy; 2025 Bright Future School. All Rights Reserved.</p>
 </footer>
</body>
</html>
```

Real-Time Project

The Full School Website Layout includes:

Navigation bar (Flexbox) ⟶ Logo + menu links.

Hero Section (Flexbox) ⟶ Welcome banner.

Main Content (Grid) ⟶ 3 cards (About, Courses, Notices).

Footer ⟶ School contact info.

Responsive design ⟶ Works on desktop and mobile.

Week 13: Final Project

Complete School Portal Website combining all:

- Registration Form
- Semantic Structure
- Media Page
- Styled with CSS selectors, box model, fonts, shadows
- Navigation bar, Flexbox, Grid layout